Removal of Specular Highlights in Endoscopic Images using 3D Gaussian Splatting

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Importance of Endoscopic 3D Reconstruction

Role of Endoscopic Imaging

- Diagnosis and treatment of lesions
- Intraoperative navigation and lesion tracking
- Construction of patient-specific digital twins

Accurate 3D scene reconstruction has become **increasingly critical** across all of applications.

Key Challenges

- Limited field of view
- Non-rigid tissue deformation
- Instrument occlusion
- Illumination variation (specular highlight) ← focus

Specular Highlight Problem

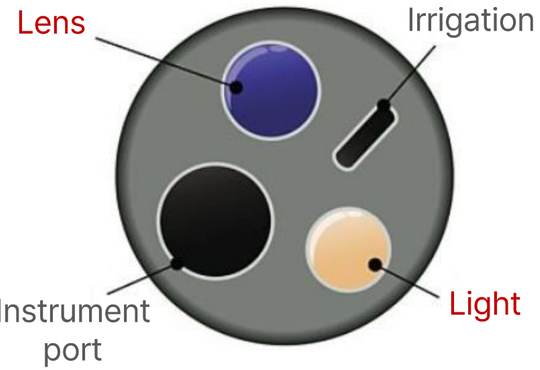


Fig 1. Configuration of an Endoscopic Instrument

Structural Root Cause

- Co-located Configuration: light source and camera are integrated at the scope tip.
- Specular Reflection: light travels and returns along the same axis, hitting the lens directly.

Critical Property: View-Dependency

- Non-Static Artifacts: highlight position and intensity shift dynamically with camera movement.
- Data Inconsistency: violation of brightness constancy across frames.

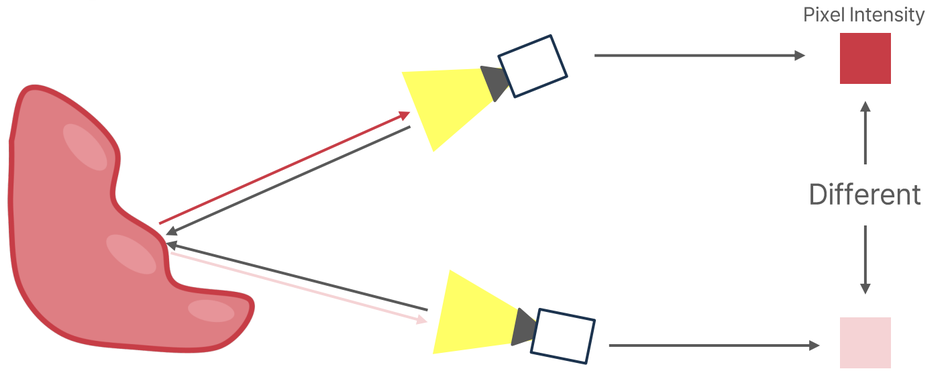


Fig 2. View-Dependent Highlight Shift

Impact of Specular Highlight

1. Tissue Appearance Distortion

- Intense specular highlights obscure the natural color and texture of the tissue, impeding clinical assessment and the accurate identification of lesions.

2. 3D Geometry Reconstruction Distortion

- Without explicit handling, view-dependent highlights corrupt Gaussian position and opacity learning
  — the 3DGS optimization absorbs specular brightness as part of the scene geometry, distorting the recovered 3D structure.

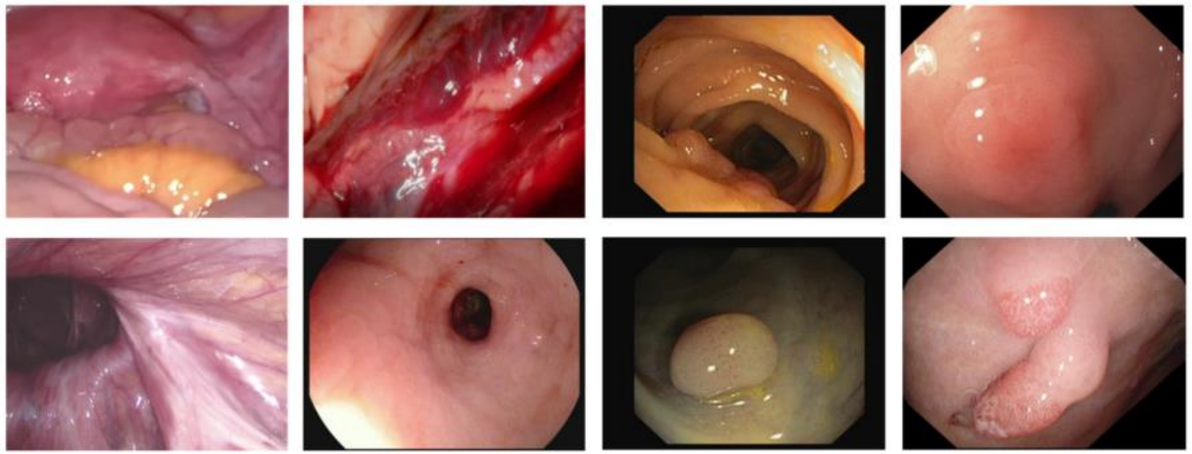


Fig 3. Endoscopy images with specular highlights[1]

[1] Nie, C., et al.: Specular reflections detection and removal for endoscopic images based on brightness classification. Sensors 23(2), 974 (2023).

Related Work

Method	Contribution	Limitation
Specular Highlight Removal in Endoscopic Images	(GAN) Specular removal from single endoscopic images [2] (Inpainting) Segmentation + attention-based partial convolution [3]	2D/single-view methods do not leverage multi-view information for joint specular removal and 3D reconstruction
3DGS for Endoscopic Scene Reconstruction	(EndoGSLAM) Real-time dense reconstruction and tracking[4] (PR-ENDO) Physically based relightable rendering[5]	3DGS-based methods focus on geometry/deformation/illumination — no explicit specular separation from input images

Our Approach

- Explicitly separating the specular component in a 3DGS-based scene representation, thereby reconstructing tissue appearance with reduced influence from specular highlights.

[2] Funke, I., et al.: Generative adversarial networks for specular highlight removal in endoscopic images. In: Medical Imaging 2018: Image-Guided Procedures, Robotic Interventions, and Modeling, SPIE, vol. 10576, pp. 1–8 (2018)
[3] Zhang, C., et al.: Specular highlight removal for endoscopic images using partial attention network. Physics in Medicine & Biology 68(22), 225009 (2023)
[4] Wang, K., et al.: EndoGSLAM: Real-Time Dense Reconstruction and Tracking in Endoscopic Surgeries Using Gaussian Splatting. In: Linguraru, M.G., et al. (eds.) MICCAI 2024, LNCS, vol. 15006, pp. 219–229. Springer, Cham (2024).
[5] Kaleta, J., et al.: Pr-endo: Physically based relightable gaussian splatting for endoscopy. In: MICCAI 2025, LNCS, vol. 16001, pp. 1–10. Springer, Cham (2025).

3DGS-based Scene Representation

- We use a set of multi-view endoscopic images $\{I_{GT}\}$ as input.
- The scene base structure is modeled using 3D Gaussian Splatting (3DGS).
- Each Gaussian primitive is parameterized by position, scale, opacity, and appearance, and is projected to each view via differentiable rendering.

- Each observed image is decomposed as:

$$I_{GT} \approx I_{base} + I_{spec}$$

- I_{base}: tissue appearance with specular removed
  I_{spec}: view-dependent specular component

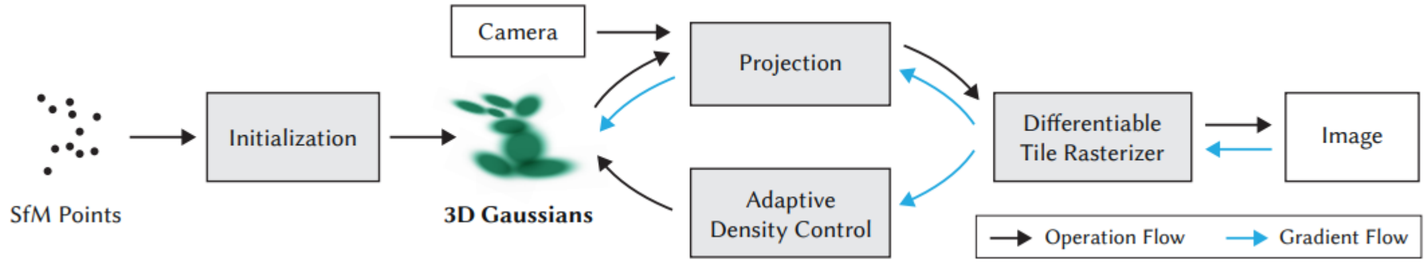


Fig 4. 3DGS Pipeline

Specular Estimation

HSV-based Prior

$P_{HSV} = ((1 - S) \cdot V)^{\gamma}$

- S (Saturation): Lower → higher specular probability
- V (Value): Brighter → higher specular probability
- γ: sensitivity control

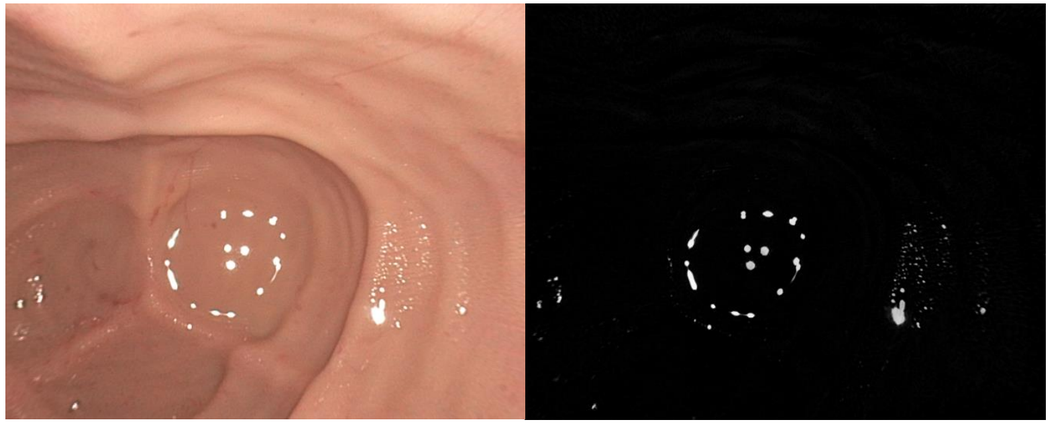


Fig 5. Input Image and estimated specular highlights

Positive Residual

$R_{spec} = ReLU(I_{GT} - I_{base})$

- Selectively identifies excess brightness unexplained by the base structure as specular candidates

Specular Loss

$L_{spec} = \left\| P_{HSV} \odot (R_{spec} - I_{spec}) \right\|_1$

- Emphasizes strongly highlighted regions.

Specular Highlight Removal

- Specular estimate may be inaccurate in strong highlight regions, so simple subtraction can leave errors in the corrected image.
- To reduce the influence of uncertain pixels, we define a pixel-wise reliability weight:

$$\omega = clamp(1 - \alpha P_{HSV}, \omega_{min}, 1)$$

- Pixels with higher specular probability receive lower weights, so more reliable observations dominate base learning.
- The base structure is optimized using the weighted reconstruction loss:

$$L_{base} = \left\| \omega \odot \left(\left(I_{GT} - I_{spec} \right) - I_{base} \right) \right\|_1$$

- The final objective is:

$$L_{total} = L_{base} + \lambda_{spec} L_{spec}$$

Qualitative Results

- Dataset: C3VD(Colonoscopy 3D Video Dataset)
- 3DGS reproduces brightness saturation and color distortion in strongly highlighted regions
- Ours removes highlights and reconstructs consistent tissue color and texture

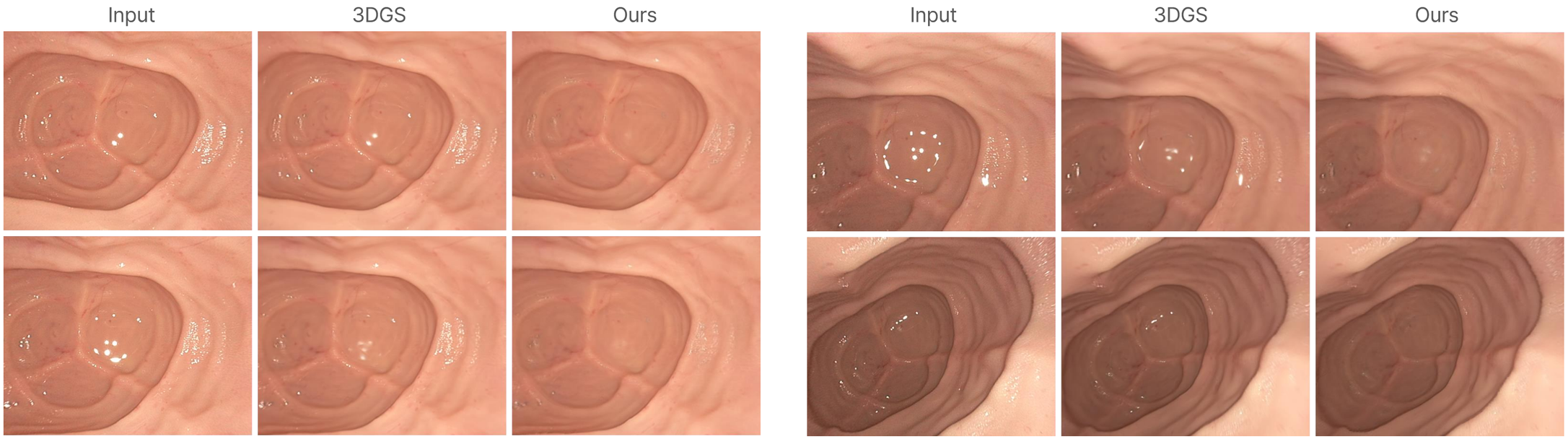


Fig 6. specular highlight removal results

Qualitative Results

- Separated specular mask consistently corresponds to highlight regions across all views

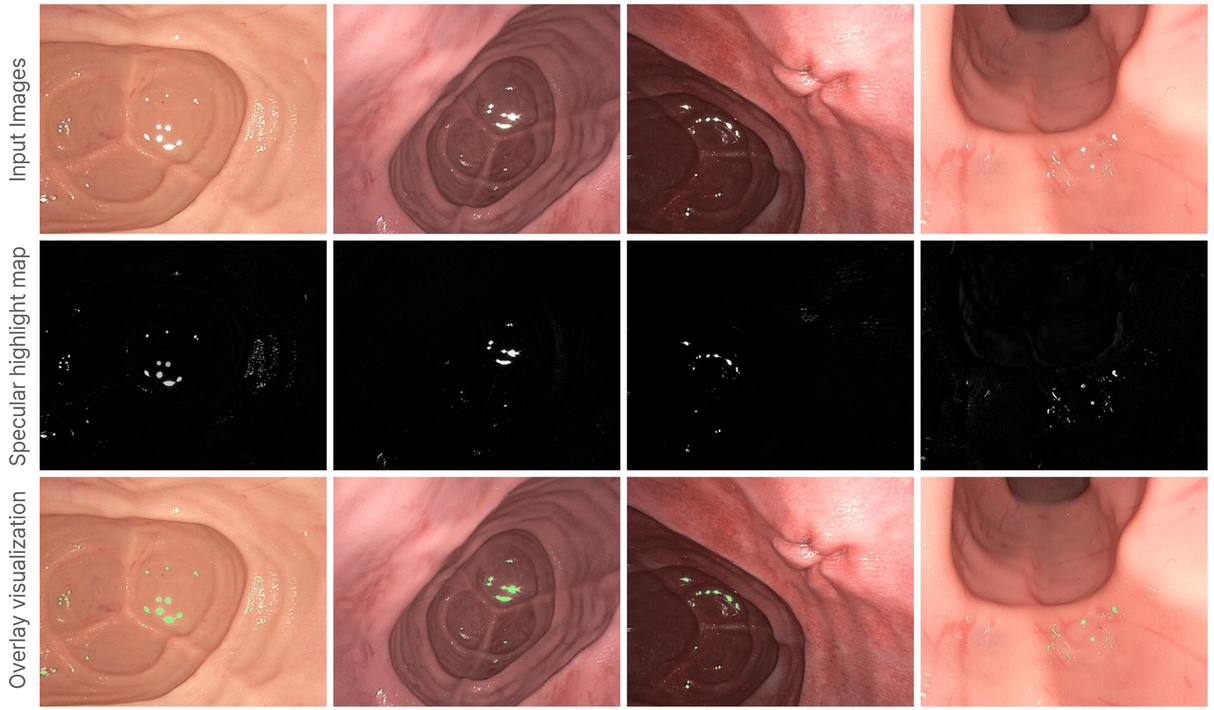


Fig 7. specular highlight estimation results

Quantitative Results

- Quantitative evaluation is performed on regions excluding specular highlight areas, enabling direct assessment of reconstructed tissue appearance quality.
- Improves appearance quality while maintaining perceptual quality (LPIPS).

Method	PSNR ↑	SSIM ↑	LPIPS ↓
3DGS	35.80	0.89	0.22
Ours	35.88	0.90	0.22

Conclusion

- A 3DGS-based framework that explicitly estimates and removes specular highlights by decomposing input images into a view-independent base and a view-dependent specular component
- HSV-based prior combined with positive residuals to selectively estimate the specular component, with decoupled optimization preventing contamination of base structure learning
- Reliability-weighted reconstruction loss that suppresses highlight uncertainty from propagating into base structure learning, improving both appearance and geometry reconstruction

Thank you